* Big-Oh Notation : Def^n :

Suppose, that an algo has complexity of

$$O(N^3)$$

Q What does it mean in simple word?

Ans

Big-Oh Notation: What does it mean?

1) It represents the Upper Bound.

In simple language, Big O provides a "guarantee" or a "ceiling." It describes the worst-case scenario for a specific algorithm, ensuring that the execution time will never grow faster than this limit as the input size N increases.

2) It describes the Growth Rate.

Meaning, if an algorithm is $O(N^3)$, the number of operations will grow in a cubic fashion relative to the input size. For example, if you double the input size ($2N$), the time taken will increase roughly eight-fold ($2^3 = 8$). It defines the mathematical relationship between the input size and time/space.

3) It acts as a "Fixed Ceiling" for a specific algorithm.

If the specific algorithm you have written is $O(N^3)$, it means that N^3 is the absolute maximum growth rate for that code.

- **Correction:** An $O(N^3)$ algorithm doesn't "change" into $O(N)$ or $O(1)$ just because the input is small; rather, Big O tells us that even in the **worst possible case**, the algorithm's complexity will **never exceed** the N^3 curve.
- It will never behave like $O(N^4)$ or $O(2^n)$. It provides a "safe zone" or an upper limit that the function's value will never cross.

* Maths :

$$f(n) = O(g(n))$$

$$\lim_{n \to \infty} = \frac{f(n)}{g(n)} < \infty$$ – {It is actually some finite value}

As we said the "Always look for worst case complexity" & "Always look at complexity for large / ∞ data".

So, Here we're applying that.

$$\lim_{n \to \infty} = \frac{f(n)}{g(n)} < \infty$$

Eg:

$$\underset{g(n)}{O(N^3)} = \underset{f(n)}{O(6N^3 + 3N + 5)}$$

$$= \lim_{n \to \infty} \frac{6N^3 + 3N + 5}{N^3}$$ - {limit when the value of n reaches / closed to ∞}.

$$= \lim_{n \to \infty} 6 + \frac{3}{N^2} + \frac{5}{N^3}$$

$$\therefore 6 + \frac{3}{\infty} + \frac{5}{\infty}$$ - {anything ÷ by $\infty = 0$}

$$= 6 + 0 + 0$$

$$= 6 < \infty$$

Hence, proved! It is a finite value because this is showing an upper bound.

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